

Star Destiny: Investigate how the Densities of Protostellar Jets affect the Efficiency of Star Formation

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Introduction

Star formation controls the evolution of galaxies, the buildup of heavy elements in the Universe, which is responsible for the origin of planetary environments in which life is possible.

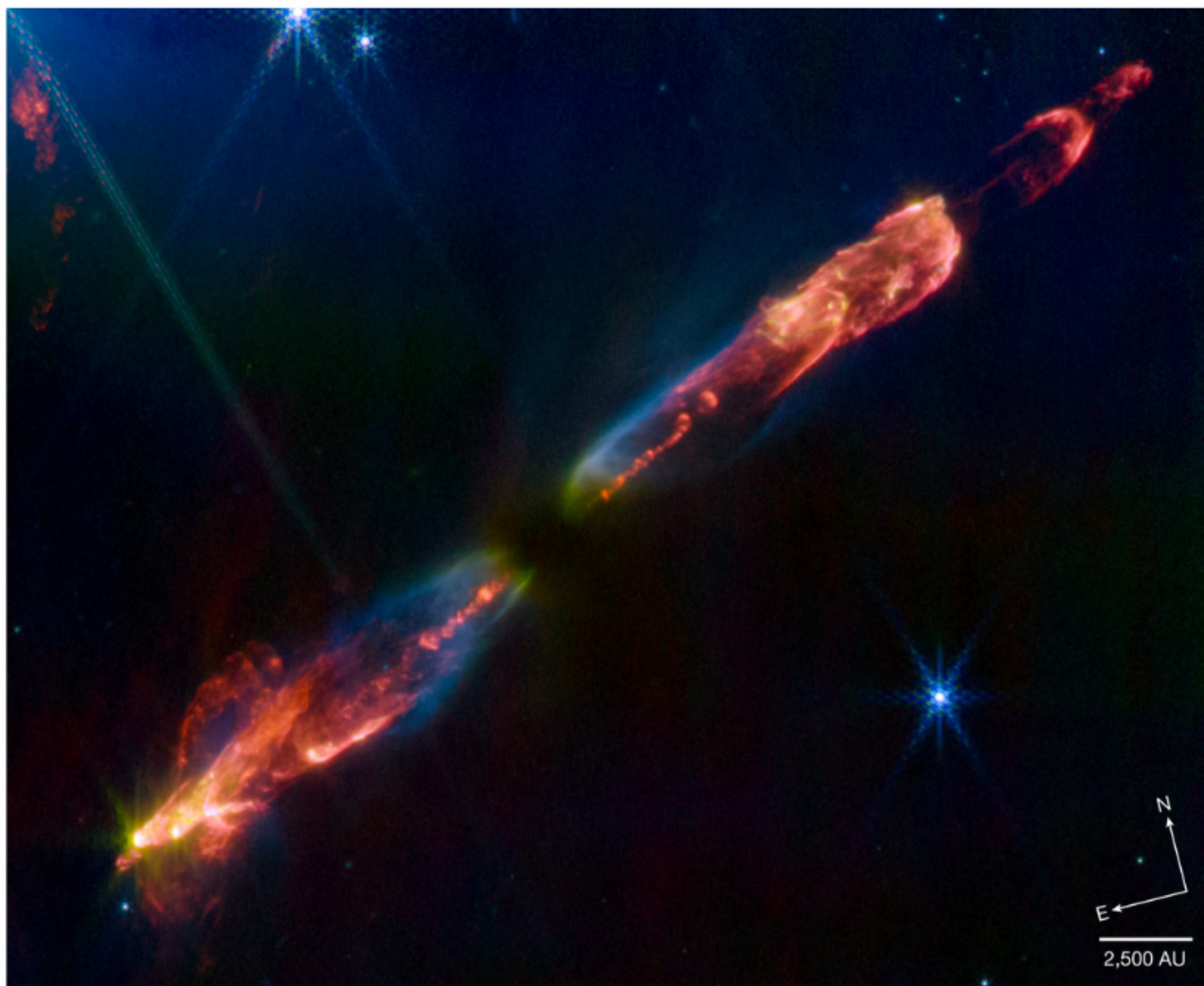


Figure 1: JWST image of HH 211 protostellar jets[1]

Protostellar Jet

- is one of the **most** essential mechanisms that protostars (new born stars) use to regulate star mass.
- high-speed, collimated streams of plasma and gas ejected from poles of protostars

Research Background

Previous studies

- indicate protostellar jets generally **lowers SFE** (star formation efficiency) by removing momentum and disrupting gas from further accretion that would otherwise increase star mass

- examined **collective** effects of jet properties, shocks, and magnetic fields on SFE, but **direct** correlation between protostellar jet density and inhibition of star formation remains **unclear**

Direct observations of star formation process are scarce because protostars are short-lived and rare.

Application

- First** to investigate the **independent** effect of protostellar jets density on star formation.
- Provide a **cost-effective (~20min/run)**, **flexible computational framework** that can be further incorporated into larger scale star formation simulations
- Provide a **quantitative prediction** that can be used to infer jet strength with SFE or estimate SFE with observed jet properties

Hypothesis

With constant momentum flux, if the **density of protostellar jet** increases, then **SFE (star formation efficiency)** will be reduced more than it does with low-density protostellar jet.

Key References

1. Ray, T. P., McCaughrean, M. J., Caratti o Garatti, A., et al. (2023). Outflows from the youngest stars are mostly molecular. Nature, 622(7981), 48–52. <https://doi.org/10.1038/s41586-023-06551-1>. 2. SWIFT Collaboration (Leiden Observatory) Sink particles: theory (GEAR subgrid model documentation). <https://swift.strw.leidenuniv.nl/docs/SubgridModels/GEAR/sinks/theory.html>. 3. Gong, H., & Ostriker, E. C. (2013). Implementation of Sink Particles in the Athena Code. The Astrophysical Journal Supplement Series, 204(1). <https://iopscience.iop.org/article/10.1088/0067-0049/204/1/8>. 4. Offner, S. S. R., & Chaban, J. (2017). Impact of Protostellar Outflows on Turbulence and Star Formation Efficiency in Magnetized Dense Cores. The Astrophysical Journal, 847, 104. <https://doi.org/10.3847/1538-4357/aa8996>. 5. Murray, D., Goyal, S., & Chang, P. (2018). The effects of protostellar jet feedback on turbulent collapse. Monthly Notices of the Royal Astronomical Society, 475(1), 1023–1035. <https://doi.org/10.1093/mnras/stx3153>. 6. Frank, A., Ray, T. P., Cabrit, S., et al. (2014). Jets and Outflows from Star to Cloud: Observations Confront Theory. In Protostars and Planets VI (pp. 451–474). University of Arizona Press. https://doi.org/10.2458/azu_uapress_9780816531240-ch020.

Algorithm Design

Jeans Normalization

$$L_J = c_s \sqrt{\frac{\pi}{G \rho_0}} = 2.814 * 10^3 \text{ AU}$$

$$t_{\text{freefall}} = \sqrt{\frac{3\pi}{32 G \rho_0}} = 6.81 \times 10^{11} \text{ s}$$

Athena++ code units are dimensionless, this helps to define physical scale of time, density, length using isothermal sound speed

Accretion

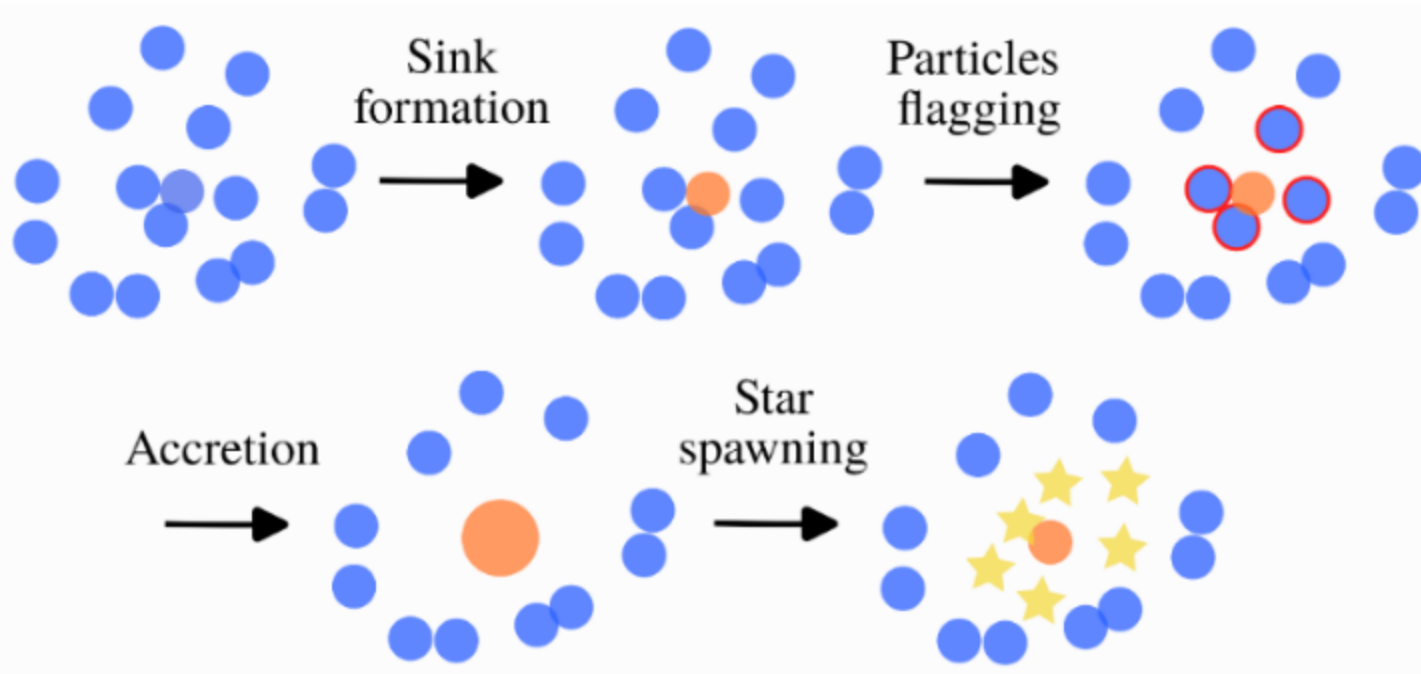


Figure 2: sink particle accretion process [2]

When the density of certain region within the sink region reach density threshold, mass of that region collapse and becomes part of a star.

Sink Gravity

$$\mathbf{a} = -\frac{G M_{\text{sink}}}{(r^2 + \epsilon^2)^{3/2}} \mathbf{r}$$

- Each cell is attracted by the point mass at the center of the sink particle
- The sink pulls more (accelerating) mass towards the center, increases center density and thus triggers accretion

Jets

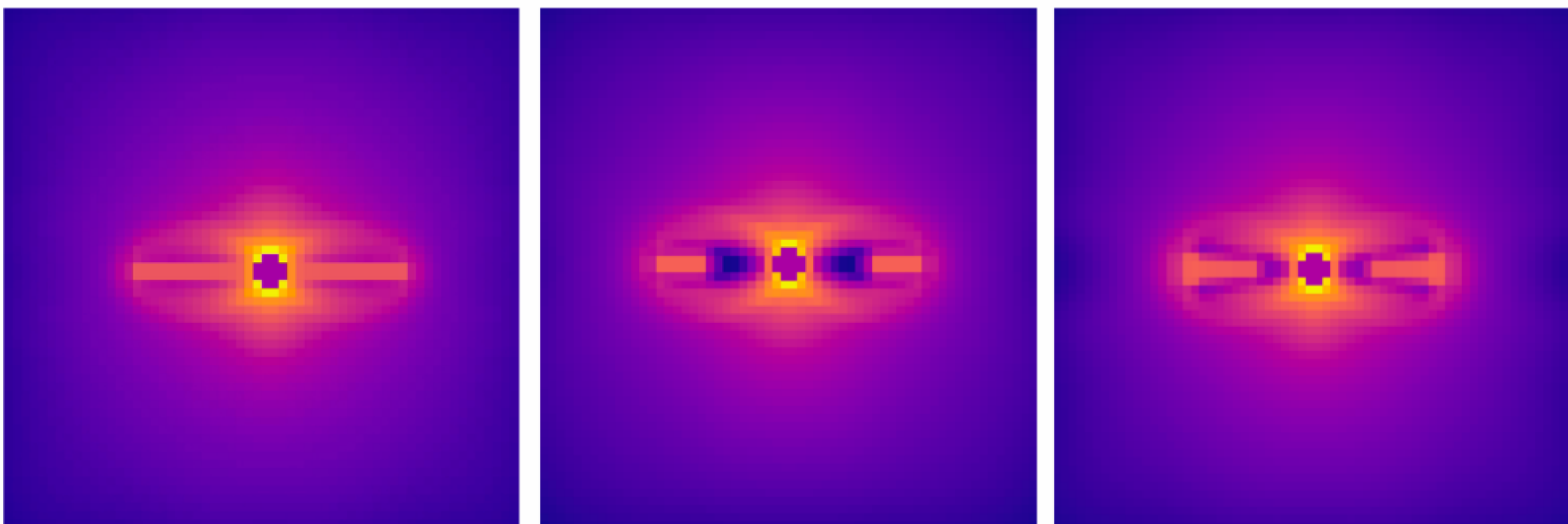
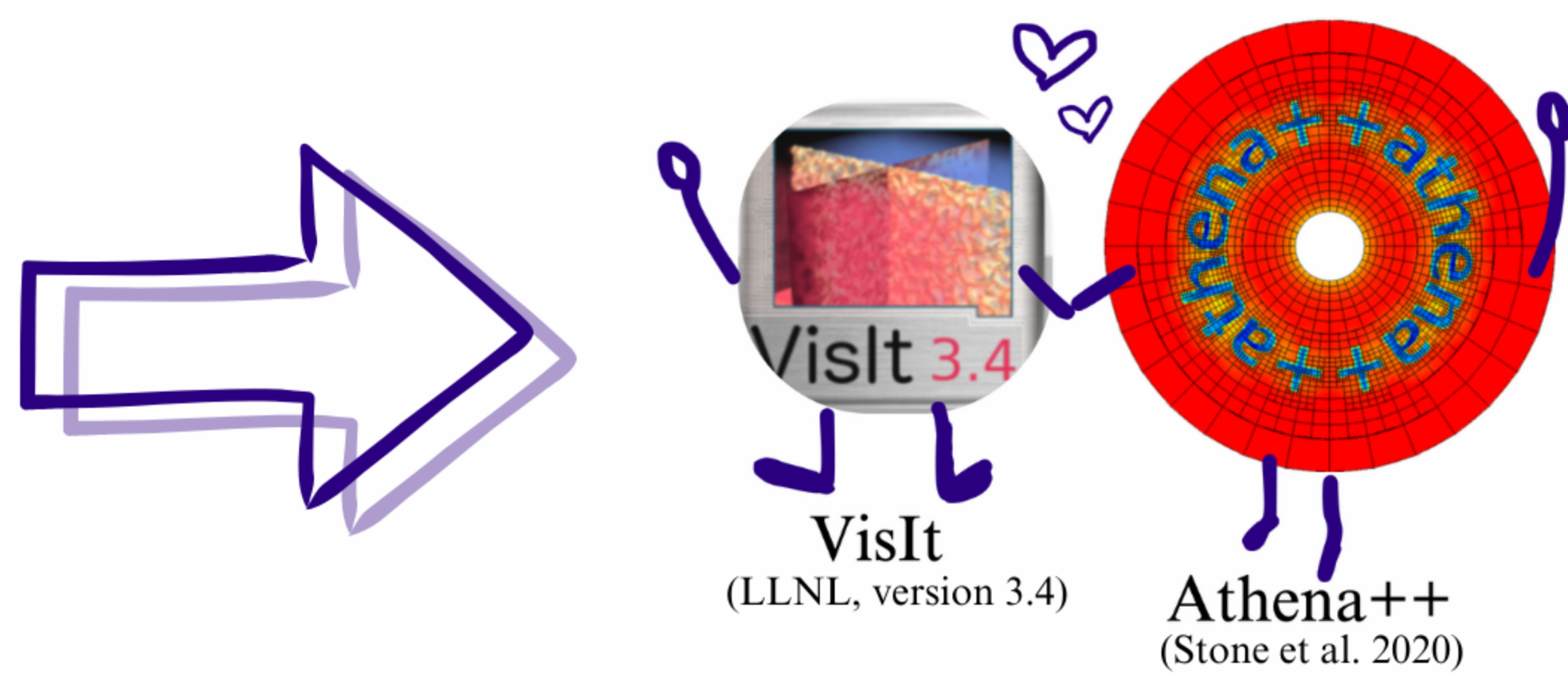


Figure 3: Three different geometries of jets, modeled by VisIt

$$\dot{\mathbf{P}}_{\text{total}} = 2 \rho_{\text{jet}} \mathbf{v}_{\text{jet}}^2 \mathbf{A}_{\text{noz}}$$

- calculate mass and corresponding velocity needed to raise nozzle region density up to ρ_{jet}
- ensure jet momentum removal rate is ~ 0.1 of accretion rate
- Inject momentum to gas in the nozzle region



Simulation Execution

- simulations were run under **8** different jet densities (1 ~ 5)
- chosen from literature review and trials (making sure no time collapsing)

Data Analysis & Visualization

$$\text{SFE}_{\text{inst}}(t) = \frac{M_{\text{sink}}(t)}{M_{\text{sink}}(t) + M_{\text{gas}}(t)}$$

- SFE_inst is recorded and output in VTK files
- analyze data with python plotting SFE_inst v.s. ρ_{jet}
- visualized with VisIt 3D density models

Conclusions

- We **verify** and **quantify** that, with constant injected momentum flux, **instantaneous SFE decreases quadratically with increasing jet density**.
- Consistent** relationship appears
 - across **all three possible jet geometries**
 - across data taken at **5 different time intervals**
- Quadratic** relationship is determined because
 - Linear** regression: **nonrandom** residuals, with a clear **quadratic** shape
 - Logarithmic and power regression residuals: **nonrandom**
 - Quartic and cubic regression: very high R^2 values ~ 0.9999 , attributed to **overfitting**; lack of precedent in astrophysics; **nonrandom** residuals
 - Quadratic regression: reasonable **R^2 values**, **random** residuals with low **R** value

Future Research

- Future **observational surveys** can be used to further constrain our simulation by comparing observed SFE with inferred jet density in star forming region
- Include **magnetic field** and **thermal feedbacks**, and measure how SFE and jet density relationship shifts relative to our current baseline.
- Add a **second** sink particle and measure whether interactions between jets can changes trend of SFE V.S. jet density.

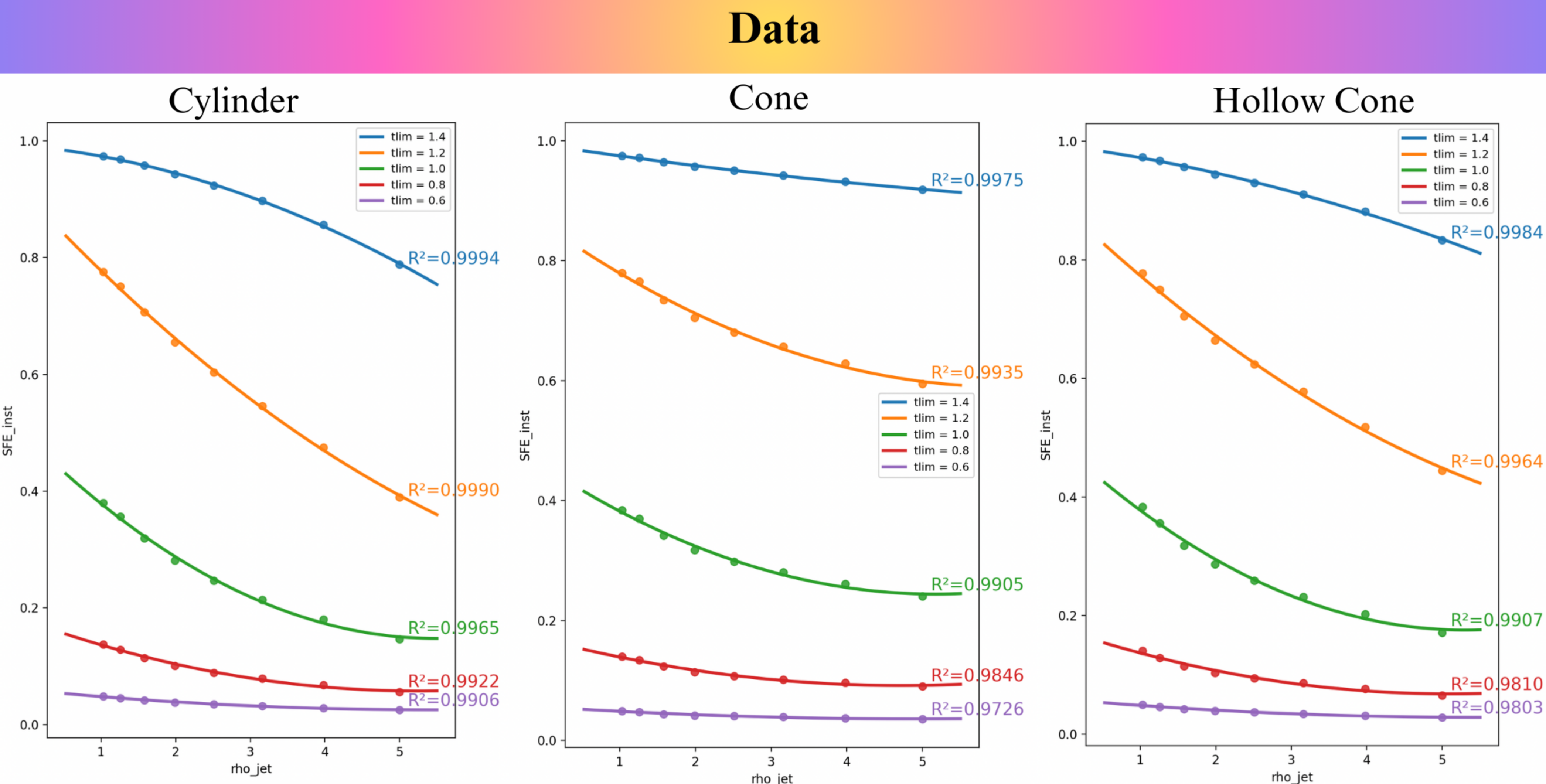


Figure 5: SFE_inst V.S. rho_jet created with python

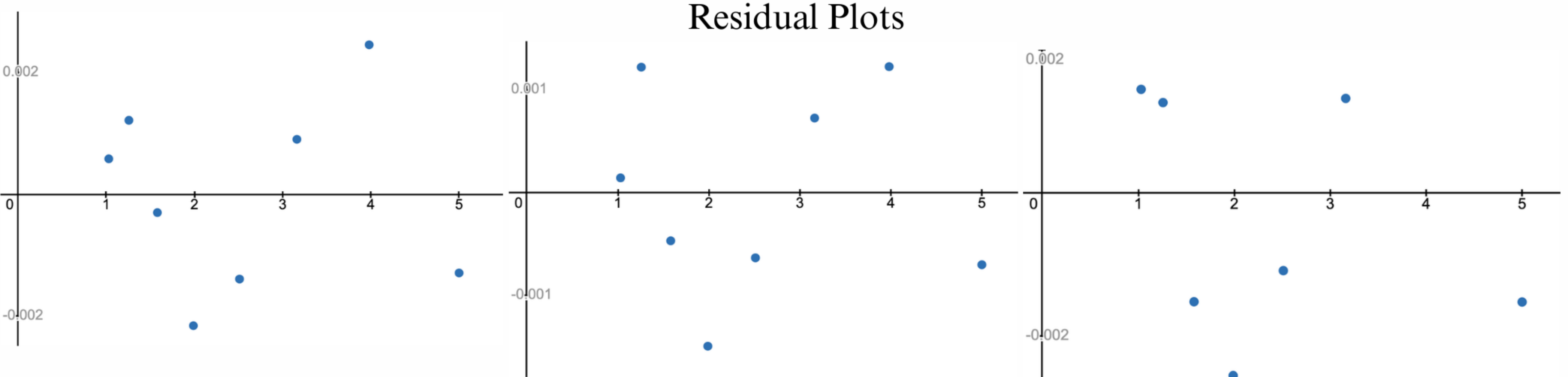
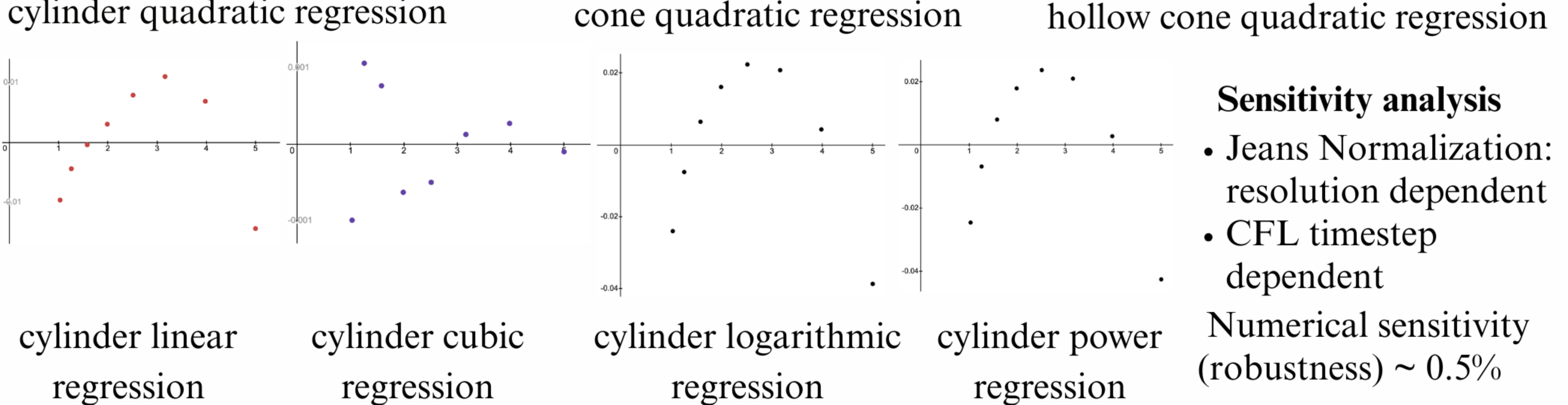


Figure 6: residual plot of different regression model created with desmos



Sensitivity analysis

- Jeans Normalization: resolution dependent
- CFL timestep dependent

Numerical sensitivity (robustness) $\sim 0.5\%$